lender can act on the information—expand lending, price risk better, cut defaults—by enough to justify paying for the data.

The same problem appears in infrastructure: a satellite can spot changes across thousands of kilometres of roads, pipelines or construction sites, but turning that insight into a recurring contract requires proving that acting on the information saves the customer more money than the cost of the data.

The ecosystem, as a result, is stuck in a chicken-and-egg loop. Customers are reluctant to build new workflows around satellite intelligence before its value is proven; but space companies need customers to generate the revenue and data needed to prove that value. Until that cycle is broken, government and foreign customers will remain the industry's deepest pockets.

To be sure, companies are waking up to the value of space data. Reliance Jio, for example, is exploring a sovereign satellite constellation; Airtel is integrating satellite connectivity into its telecom business; and banks and insurers are beginning to use satellite intelligence for agricultural lending and risk. However, these markets are still narrow, concentrated and emerging.

This creates a second bottleneck: growth capital. Start-ups need larger cheques as they move from prototypes to production, yet the funding pattern in Indian spacetech has been dominated by early-stage capital. Only two commercial space companies—Skyroot Aerospace and Pixxel—have been able to raise more than $100 million. Experts say capital is difficult to raise without visibility on future revenues. And revenue visibility, in turn, depends on having anchor customers.

Anirudh Sharma, founder and CEO of domain awareness startup Digantara, points to SpaceX, or Space Exploration Technologies Corp., the aerospace company founded by Elon Musk.

"Even before SpaceX launched the Falcon, they had contracts from NASA, from defence. We don't yet have that established mechanism to win contracts at a developmental stage," he says. "Today, all the development needs to be funded internally."

The government has become an "anchor customer", he says, but it is not yet consistently acting as a developmental partner that commits to buying a technology before it is proven. When a government commits to buying 30 or 40 systems once the first one works, a company can build its factory, hire engineers and raise capital against plausible future revenue. However, if there is commitment for only one system, the economics don't work. "We don't need grants at this point in time," Sharma says. "We need orders, contracts to develop capacity for the country."



Skyroot is using Siemens' Polarion to unify its software and systems development processes

In some sectors, such as defence, it is already underway. In 2024, the Cabinet Committee on Security headed by PM Modi approved ₹26,968 crore for Phase III of the space-based surveillance mission, aimed at improving land and maritime domain awareness for civilian and military applications. Under the programme, a constellation of 52 satellites is scheduled to be deployed by 2029, of which 31 are to be built by private space companies.

Then, in January this year, IN-SPACe, the Indian National Space Promotion and Authorisation Centre, an interface between the government and private space companies, signed an agreement with Allied Orbits,